

1. Why Prior Knowledge Matters?

- Real-world learning is more effective when the agent has prior knowledge about the domain.
- Prior knowledge constrains the hypothesis space and biases the search toward plausible hypotheses.
- Helps learn from fewer examples, generalize better, and avoid implausible hypotheses.
- Encodes useful structure, relations, constraints, and abstractions about the world.

Key Idea: Learning is more sample-efficient, more accurate, and more robust when combined with prior knowledge.

Example: A child learns what a "bird" is faster if it knows birds typically have wings, can fly, and lay eggs.

4. Explanation-Based Learning (EBL)

- Uses a domain theory to explain observed training examples.
- Generalizes the explanation to a more general rule (without needing many examples).
- Learns by deduction, not by induction.
- Produces a rule that covers the training example and is entailed by the domain theory.

Algorithm (EBL)

1. Input: training example e , domain theory D .
2. Find an explanation of e using D (derive why e is true).
3. Generalize the explanation by removing example-specific details.
4. Add the generalized rule to the hypothesis.
5. Use the new rule for future predictions.

Example: Learning the concept "grandparent"

- Training example: grandparent(John, Mary)
- Domain theory: parent and ancestor relations.
- Explanation: John is a grandparent because John is a parent of x and x is a parent of Mary.
- Learned rule: $\text{grandparent}(X, Z) :- \text{parent}(X, Y) \wedge \text{parent}(Y, Z).$

7. What Constitutes a Domain Theory?

- A domain theory is a set of background knowledge about the target concept and the domain in which it is defined.
- Typically expressed in a formal language such as propositional or first-order logic.
- Includes definitions, relations, constraints, and general rules about the world.
- Used both to explain examples (EBL) and to constrain/generalize learning.

Examples of domain theory components:

- Definitions (e.g., ancestor, sibling, parent)
- Taxonomies (e.g., bird $\Rightarrow$ animal)
- Physical knowledge (e.g., objects fall)
- Constraints (e.g., mutually exclusive properties)
- Causal knowledge (e.g., rain causes wet ground)

2. Using Background Knowledge to Improve Learning

- Background knowledge can be used to:
 - Restrict the hypothesis space (eliminate implausible hypotheses).
 - Provide useful features or abstractions.
 - Suggest the form of hypotheses (bias).
 - Help interpret and explain training data.
 - Allow generalization from very few examples.
 - Improve sample efficiency and reduce overfitting.

Example: Recognition with background knowledge

To recognize a zebra, prior knowledge that zebras are a type of horse with black-and-white stripes helps the learner focus on relevant features and avoid confusing it with other animals.

5. Example of EBL: Blocks World

- Domain theory: definitions of on, clear, block, and rules about how actions change states.
- Training example: $\text{On}(A, B).$
- Explanation: from the domain theory, $\text{On}(A, B)$ is true because A was moved onto B and B was clear.
- Generalized rule: $\text{On}(x, y)$ can produce supports (x, y) for the planning domain.
- Learned rule can be used to recognize similar situations with new objects.

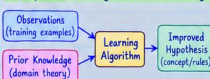
8. Learning from Observations with Prior Knowledge

- Combine observed examples with background knowledge to learn a better hypothesis.
- The learner uses the domain theory to interpret observations and infer general rules.
- Can fill in missing information and derive hidden structure not directly observed.

Example: Learning with noisy or partial data

- Observations: a few examples of animals with wings and that can fly.
- Prior knowledge: birds are animals, typically have wings, can fly, and lay eggs.
- Result: learn a more accurate and general concept of bird from fewer examples.

9. Pipeline: Learning with Prior Knowledge



Prior knowledge guides the learner to consider better hypotheses, leading to faster and more accurate learning.

3. The Problem of Integrating Learning and Reasoning

- Real agents must both learn from data and use their knowledge to reason and act.
- Integrating learning and reasoning is difficult because:
 - The learned knowledge must be consistent with existing background knowledge.
 - Learning may introduce new symbols, relations, or concepts.
 - Reasoning with an incorrect learned hypothesis can lead to wrong inferences.
 - The representation language must be rich enough for both learning and reasoning.
 - Computational complexity can be high.

Key Challenge: How to modify and extend the agent's knowledge base based on experience while maintaining consistency and enabling effective reasoning.

6. Forms of Prior Knowledge in Learning

- Knowledge about objects and categories (e.g., hierarchies, taxonomies).
- Relations and constraints (e.g., part-of, temporal, spatial).
- Domain-specific rules (e.g., physics, causality, default assumptions).
- Biases about useful hypothesis forms.
- Abstractions that group similar instances.
- Structured representations (e.g., frames, logic rules, probabilistic models).

Example: Using constraints

- Domain knowledge: a block can be on at most one block, and support requires contact.
- These constraints help eliminate impossible hypotheses when learning from observations.

10. Summary Example: Animal Classification

- Observations: labeled and unlabeled images of animals.
- Prior knowledge: taxonomic hierarchy (birds are animals), properties (feathers, wings, lay eggs), and constraints.
- Learning result: a hypothesis for the concept of bird that generalizes beyond the training examples using the provided knowledge.

11. Key Takeaways

1. Prior knowledge significantly improves learning performance.
2. Background knowledge constrains the hypothesis space and guides generalization.
3. Explanation-based learning uses domain theory to learn from few examples.
4. Integrating learning and reasoning is challenging but essential for intelligent agents.
5. Domain theory includes definitions, relations, constraints, and general rules.
6. Learning from observations plus prior knowledge yields more accurate, general, and sample-efficient hypotheses.

1. What is Explanation-Based Learning?

- Explanation-Based Learning (EBL) learns by using a domain theory to explain why a training example satisfies a target concept, and then generalizes the explanation into an operational rule.
- EBL is a form of analytical learning: it does not search through many hypotheses, but uses an explanation to focus on a relevant rule.

Key Idea: If we can explain why an example is a positive instance using background knowledge, we can generalize that explanation to new, similar instances.

Example: Explain why a particular block stack is safe to stack, then generalize the explanation to a rule for all similar stacks.

4. Generalizing the Explanation

- The proof for a specific example often contains details that are specific to that example (e.g., object names, numeric values).
- We generalize the proof by removing example-specific details and retaining only the relevant, essential conditions.
- The result is a more general rule that applies to a wider class of instances.

Generalization process:

- Identify which parts of the proof are essential for the target concept.
- Replace example-specific constants with variables.
- Remove irrelevant details.
- Simplify to obtain a concise rule.

7. Explanation for the Example

- Using the domain theory, we can derive:
 - b is on the table $\rightarrow$ b is stable.
 - a is smaller than b.
 - a is lighter than b.
 - Therefore, stack(a, b) is safe.

Proof/Derivation (sketch):



10. Benefits of Explanation-Based Learning

- Learns accurate and general rules from a single or few examples.
- Uses existing domain knowledge to focus learning and avoid irrelevant hypotheses.
- Learned rules are operational and efficient to apply.
- Particularly effective when examples are sparse and domain theory is strong.
- Learns deeper, more meaningful knowledge than rote memorization.

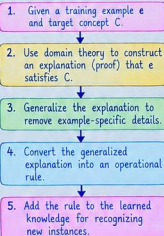
2. Operationality of Learned Knowledge

- The learned knowledge must be operational, i.e., directly usable for recognizing or generating new instances.
- The explanation is turned into an operational rule that can be applied efficiently, typically as a set of conditions.
- Operational rules are usually expressed as if-then rules or decision procedures.

Operational Rule (general form):

IF conditions derived from explanation
THEN example is an instance of the target concept.

5. EBL Workflow



8. Generalized Explanation (Rule)

- Remove example-specific details (a, b) and keep the essential conditions.
- The generalized explanation becomes a rule:

IF Small(x, y) $\wedge$ Light(x, y) $\wedge$ Stable(y)
THEN Safe(x, y)

- This operational rule can now be used to recognize new safe stacks, such as stack(c, d) where c is a small, light block and d is stable.

11. Limitations of EBL

- Requires a correct and sufficiently complete domain theory.
- Explanation must exist; cannot learn if the example cannot be explained.
- Learns only from positive examples (traditionally).
- May overgeneralize if domain theory is incorrect or too weak.
- Not effective in noisy or uncertain domains.
- Does not learn new domain knowledge (only uses existing knowledge).

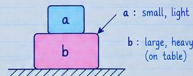
3. Using Domain Theory to Explain

- A domain theory provides background knowledge about how the world works (e.g., physics, geometry, causal relations).
- To explain an example, we construct a proof (or derivation) showing that the example satisfies the target concept, using the domain theory.
- The explanation is usually a proof tree or chain of reasoning.

Explanation: A demonstration, using the domain theory, of why the training example is a positive instance of the target concept.

6. Worked Example: Safe-to-Stack

- Target concept: Safe(x) — block x can be placed on block y without the stack falling.
- Domain theory (simplified):
 - A block is stable if it is on the table or on a stable block.
 - A block can be placed on another block if it is smaller and lighter.
- Training example: stack(a, b) where a is a small, light block and b is a large, heavy block on the table.



9. Applying the Learned Rule

- The learned rule is operational and can be applied directly to new instances.
- Example: stack(c, d)
 - Small(c, d): true
 - Light(c, d): true
 - Stable(d): true (d is on table)
 - Therefore, Safe(c, d): true

New instance:



12. EBL Assumptions (Summary)

- A domain theory is available and (mostly) correct.
- Training examples are positive instances of the target concept.
- Each training example can be explained using the domain theory.
- The explanation can be generalized to produce an operational rule.
- The target concept is well-defined and expressible in the domain theory.
- The learned rule is efficient to apply.

1. Why Knowledge Matters in Learning

- Using background knowledge can make learning more efficient and accurate.
- Knowledge helps focus attention on relevant features and rule out implausible hypotheses.
- It guides the choice of representation and the search through hypothesis space.
- Learning with knowledge can generalize better from fewer examples.

Key Idea: Learning is not just from raw data; it is learning with the right knowledge in the right form.

4. Choosing the Right Representation

- The choice of representation determines the hypothesis space and what can be learned.
- A poor representation may be too weak (cannot express the target concept) or too strong (search becomes intractable).
- Background knowledge helps us design a representation that matches the problem and uses the right features and structure.

Example: For handwritten digit recognition, using pixel features is possible but high-dimensional; knowledge about strokes and shapes can lead to a better representation.

7. Deductive vs. Inductive Learning

- Deductive learning uses logical inference from general knowledge to derive specific conclusions (explanations).
- Inductive learning uses examples to infer general rules.
- Deduction can produce highly specific hypotheses but needs correct background knowledge.
- Induction is more flexible and can learn from noisy or incomplete data.
- Combining both leverages the strengths of each approach.

10. Compare and Contrast

Aspect	Deductive Learning (Explanation-Based)	Inductive Learning
Source of knowledge	Background knowledge and logical inference	Training examples
Type of inference	Deductive (general → specific)	Inductive (specific → general)
Resulting hypothesis	Specific, justified by explanation	General, from observed examples
Data needed	Few examples (sometimes one)	Many examples
Handles noisy data	Not well	Better (with suitable algorithm)
Strengths	Uses domain knowledge, efficient, interpretable	Flexible, can learn from data without prior knowledge
Weaknesses	Requires correct background knowledge; may be too specific	May need many examples; can be misled by irrelevant features

2. Relevance-Based Learning

- Not all features in the input are relevant to the target concept.
- Relevance-based learning tries to focus on the relevant features and ignore irrelevant ones.
- This reduces the size of the hypothesis space and improves generalization.
- Relevant features are often determined using prior knowledge about the domain.

Example: When learning to play tennis, the feature "Outlook" (sunny, overcast, rain) is relevant, while "Date" or "Player name" may be irrelevant.

5. Combining Explanation-Based and Inductive Learning (FOCL)

- FOCL (First-Order Combined Learner) combines explanation-based learning (EBL) and inductive learning (IL).
- Uses background knowledge to explain training examples (like EBL).
- Generalizes these explanations to learn a more general hypothesis.
- Can learn first-order rules and handle noisy data better than pure EBL.

Example: In a blocks-world, if background knowledge explains why a block is movable in a training example, FOCL can generalize this explanation to learn a general rule.

8. How Background Knowledge Constrains the Hypothesis Space

- Background knowledge rules out implausible hypotheses, reducing the search space.
- It restricts the form of hypotheses (e.g., allowed predicates, relations, or functional forms).
- It can add constraints on feature values and structure (e.g., domain rules, type information).
- A smaller, constrained hypothesis space leads to faster and more accurate learning.

Example: If we know that "birds can fly", we do not consider hypotheses where a bird cannot fly (unless exceptions are allowed).

3. Role of Relevant Features

- Relevant features capture important aspects of the target concept.
- Irrelevant or noisy features can mislead the learning algorithm and cause overfitting.
- Feature selection (using knowledge) can improve learning speed and accuracy.
- Domain knowledge can suggest which features are likely to be relevant.

Example: In medical diagnosis, symptoms like fever and cough are relevant features, while the patient's ID number is irrelevant.

6. Learning with Prior Knowledge vs. Pure Induction

- Pure induction learns from examples without using prior knowledge.
- Learning with prior knowledge uses background knowledge to bias and guide the search.
- With prior knowledge, fewer examples are needed, learning is faster, and generalization is better.
- Prior knowledge can also prevent learning incorrect or implausible hypotheses.

Example: A child learning animals uses prior knowledge (e.g., "animals usually have four legs") to generalize from few examples.

9. Illustrative Example

Learning to predict if a person will play tennis using background knowledge.

Background knowledge:

- Weather conditions (Outlook, Temperature, Humidity, Windy) are relevant.
- PlayerName and Date are irrelevant.
- If Outlook = Overcast then PlayTennis = Yes.

Using the knowledge, the hypothesis space ignores irrelevant features:

Full feature set	Use background knowledge	Relevant features
Outlook, Temperature, Humidity, Windy, PlayerName, Date (large hypothesis space)	→	Outlook, Temperature, Humidity, Windy (smaller hypothesis space)

11. Key Takeaways

- Relevance-based learning focuses on relevant features to improve learning.
- Choosing the right representation is crucial.
- FOCL combines explanation-based and inductive learning.
- Prior knowledge helps learning by reducing the hypothesis space.
- Deductive and inductive learning are complementary.
- Background knowledge constrains the hypothesis space and improves generalization.
- Learning with knowledge can be faster, more accurate, and require fewer examples.

1. Knowledge-Based Inductive Learning

- Uses background knowledge to guide inductive learning.
- Searches only hypotheses consistent with domain knowledge.
- Can greatly reduce the hypothesis space and training data needed.
- Combines knowledge representation with machine learning.

Key Idea: Use prior knowledge to bias the search toward plausible hypotheses, making learning faster and more accurate.

Example: Learn to identify animals using biological knowledge (e.g., mammals have hair, birds have wings).

2. Analytical Learning

- Derives learned knowledge by logical inference from background knowledge and examples.
- Does not search a large hypothesis space; instead, it analyzes examples to form rules or explanations.
- Often used with logical representations (e.g., first-order logic).
- Can learn generalized rules from few examples if domain theory is strong.

Example: From examples of birds and the rule "birds typically fly", learn a rule about flying animals.

3. Prior Knowledge Can Be Incomplete or Incorrect

- Domain knowledge may be incomplete (missing important facts).
- It may also be incorrect or contain false assumptions.
- Learning must be robust to such errors.
- Incorrect knowledge can mislead the search and lead to wrong hypotheses.

Example: A medical domain theory might incorrectly state that all fevers are caused by viral infections, leading to wrong diagnoses.

Note: Learning systems should be able to revise or ignore incorrect knowledge when contradicted by data.

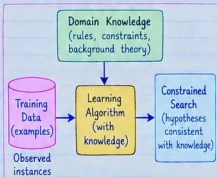
4. Dealing with Imperfect Domain Theories

- Use probabilistic or weighted knowledge (rather than absolute rules).
- Allow exceptions and uncertainty in domain knowledge.
- Learn to refine the domain theory from data.
- Use methods such as:
 - Probabilistic logical models (e.g., Markov Logic Networks).
 - Default reasoning with exceptions.
 - Learning and revising domain rules.

Example: In a medical domain, "birds fly" can be a default rule with exceptions (e.g., penguins). The system can learn exceptions from data.

5. Data + Knowledge = Constrained Search

- Prior knowledge constrains the hypothesis space to consistent hypotheses.
- Learning algorithm searches only hypotheses that satisfy the domain knowledge.
- Reduces search space and data requirements.



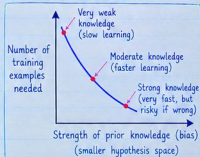
6. Benefits and Risks of Strong Background Knowledge

Benefits (Pros)	Risks (Cons)
Faster learning (smaller search space)	Incorrect knowledge can mislead learning
Needs fewer training examples	May prevent learning the correct concept (if too restrictive)
Can learn more complex concepts	Bias may lead to overconfidence
Can generalize better with sparse data	Maintenance of domain knowledge is difficult

Key Point: Good background knowledge can improve learning, but wrong or overly strong knowledge can hurt performance.

7. Learning Speed and Bias

- Bias from prior knowledge reduces the size of the hypothesis space.
- Smaller hypothesis space → faster learning (fewer examples and less computation).
- A stronger bias can lead to better generalization when the bias is correct.
- If the bias is wrong, learning can be fast but converge to an incorrect hypothesis.
- There is a trade-off between learning speed and the risk of bias error.



8. Examples

- **Medical diagnosis:** use knowledge of diseases, symptoms, and causal relations.
- **Natural language learning:** use grammar and semantic knowledge.
- **Perception:** use physical knowledge about the world (e.g., objects are persistent).
- **Game playing:** use domain knowledge about game rules and strategies.
- **Robotics:** use physical laws and constraints.

9. Limitations

- Difficult to obtain correct and complete domain knowledge.
- Domain knowledge is often uncertain and may change.
- Strong bias can prevent learning new or unexpected concepts.
- Conflicts between domain knowledge and observed data must be resolved.
- Knowledge-engineering effort can be costly.
- Not all domains have well-understood theories (e.g., complex real-world tasks).

10. Key Takeaways

1. Prior knowledge guides and constrains inductive learning (KBIL).
2. Analytical learning uses logical inference.
3. Domain knowledge can be incomplete or incorrect.
4. Learning must handle imperfect knowledge.
5. Good knowledge improves speed and generalization but can be risky.
6. There is a trade-off between learning speed and bias.
7. Works well in many domains but has practical limitations.

“Background knowledge can make learning faster and more accurate, but it must be used carefully.”

1. Key Definitions

Prior Knowledge

- Knowledge that the learner already has about the domain.
- Used to constrain and guide the search for the hypothesis.

Domain Theory

- A (typically causal) theory of how the target concept depends on other features.
- Provides a structured explanation of training examples.

Explanation-Based Learning (EBL)

- Uses a domain theory to explain a training example and generalize a more general rule that covers the example.
- Learns by explanation rather than by generalizing from many examples.

Operationality

- A concept or rule is operational if it can be effectively computed and applied by the learner (i.e., testable/usable in practice).
- Rules must be expressed in operational terms to be useful.

Relevance

- A feature or condition is relevant if it causally or reliably affects the target concept.
- Irrelevant details should be ignored to avoid overfitting and improve generalization.

Inductive Bias

- The set of assumptions a learner uses to prefer one hypothesis over another.
- Prior knowledge (including domain theory) provides an inductive bias, restricting the hypothesis space to plausible hypotheses.

6. Example: EBL in Practice

Scenario: A robot blocks world. Domain theory T:
A block will fall if it is unsupported.

Training example: Block A falls.

- Explanation using T:
 - A is a block.
 - A is not on any other block.
 - Therefore, A will fall (from T).
- Generalize:
 - Remove details specific to A (e.g., color, size).
 - Keep relevant condition: a block with nothing underneath will fall.
- Learned rule (operational):

If a block is not on another block, then it will fall.

2. Pure Inductive Learning vs. Knowledge-Guided Learning

Aspect	Pure Inductive Learning	Knowledge-Guided Learning (e.g. EBL)
Source of knowledge	Only training data	Training data + domain knowledge
Search space	Typically very large	Restricted using domain theory
Sample efficiency	May require many examples	Can learn from few examples
Type of bias	Generic inductive bias (e.g. Occam's razor)	Structured bias from domain theory
Learned rules	Generalize from observed patterns	Derived by explanation, typically more general and useful
Robustness to noise	More sensitive to irrelevant features	Theory helps focus on relevant features
Typical use	When no domain knowledge is available	When a domain theory is available and reasonably correct

5. Common Assumptions and Limitations

Common Assumptions

- A reasonably correct domain theory is available.
- Training examples are (mostly) correct positive examples.
- The domain theory can explain the examples.
- Explanations can be generalized by removing irrelevant details.
- Learned rules are operational and computable.

Limitations

- Requires a correct and sufficiently rich domain theory (not always available).
- If the domain theory is wrong or incomplete, learned rules can be incorrect.
- Sensitive to errors in the theory or examples.
- May not work well when the target concept cannot be explained by the theory.
- Less effective in noisy or real-world domains with complex, uncertain knowledge.

7. Likely Exam Questions

- Define explanation-based learning (EBL). How does it differ from purely inductive learning?
- What is a domain theory? Give an example and explain its role in EBL.
- What is inductive bias? How does prior knowledge provide an inductive bias?
- Explain the EBL algorithm with its main steps.
- Discuss the assumptions and limitations of EBL. When is it most useful?
- Compare pure inductive learning with knowledge-guided learning.
- What does it mean for a concept to be operational? Why is this important?

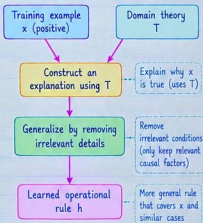
3. Explanation-Based Learning Process

EBL (Sketched Algorithm)

- Input: training example x and domain theory T .
- Explain: use T to construct an explanation of why x is a positive example.
- Generalize: remove details that are irrelevant to the explanation, producing a more general rule h that still covers x .
- Operationalize: ensure the rule h is operational (can be computed and applied).
- Add: add h to the hypothesis set.
- Use: learned rule h to classify future examples.

Key idea: EBL converts a specific example into a general rule by proving (explaining) why the example must be true given the domain theory.

4. Visual Summary: EBL



8. Key Takeaways

- Prior knowledge can greatly improve learning by constraining the hypothesis space.
- A domain theory explains how the target concept depends on other features.
- Explanation-based learning learns by explanation: explain an example, then generalize the explanation.
- EBL can learn useful, general rules from few examples.
- Learned rules must be operational and based on relevant features.
- Knowledge-guided learning is more data-efficient than pure inductive learning.
- EBL relies on a correct domain theory and can fail if the theory is wrong.
- Prior knowledge provides an inductive bias that focuses learning on plausible hypotheses.